

Hands-on Exercise 1

AUTHOR
Ye Zhili

Getting Started

Install and launching R packages

The code chunk below uses `p_load()` of `pacman` package to check if tidyverse packages are installed in the computer. If they are, then they will be launched into R.

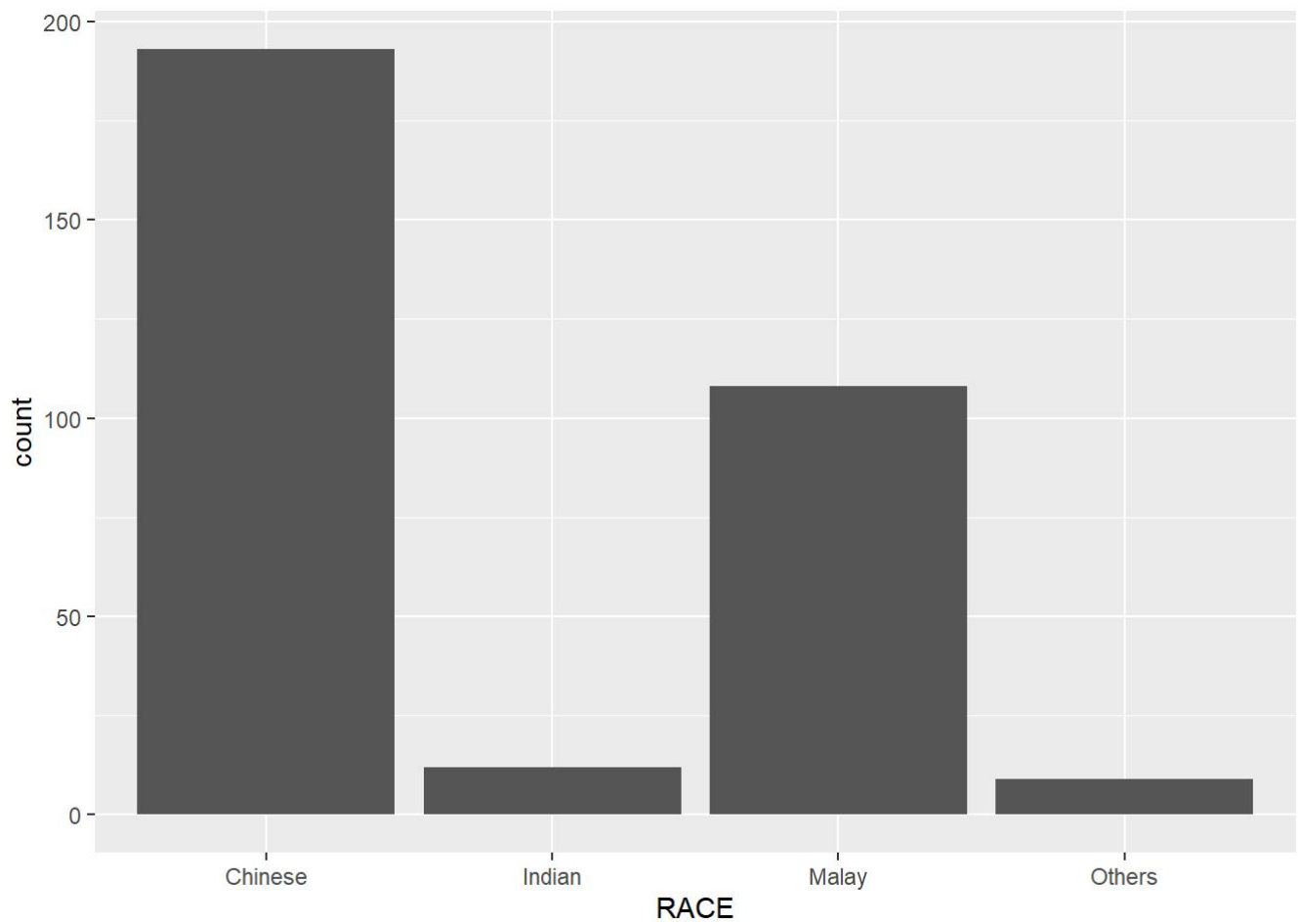
```
pacman::p_load(tidyverse)
```

Importing the data

```
exam_data <- read_csv("data/Exam_data.csv", show_col_types = FALSE)
```

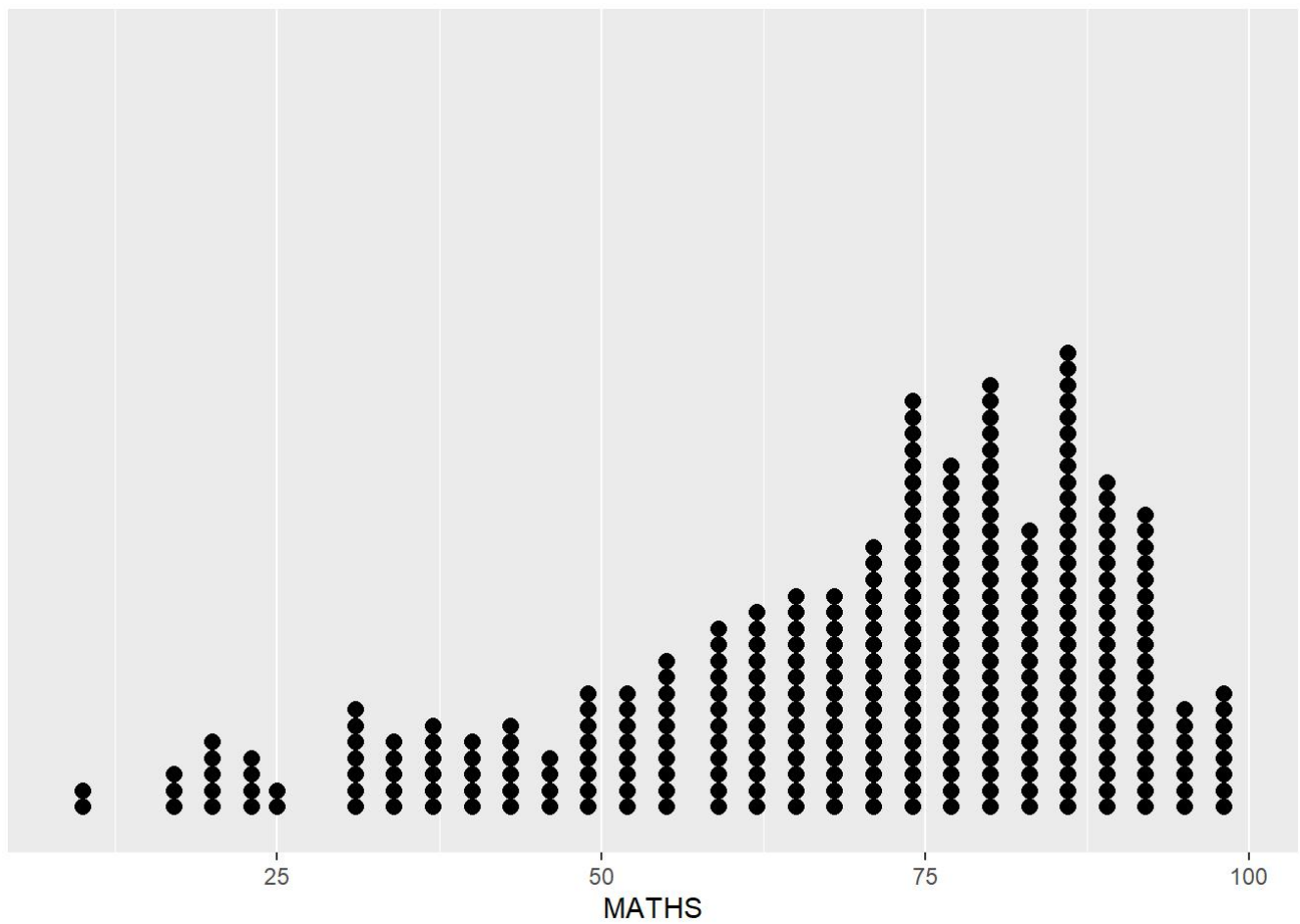
Plotting a simple bar chart

```
ggplot(data=exam_data, aes(x=RACE))+geom_bar()
```



Dotplot

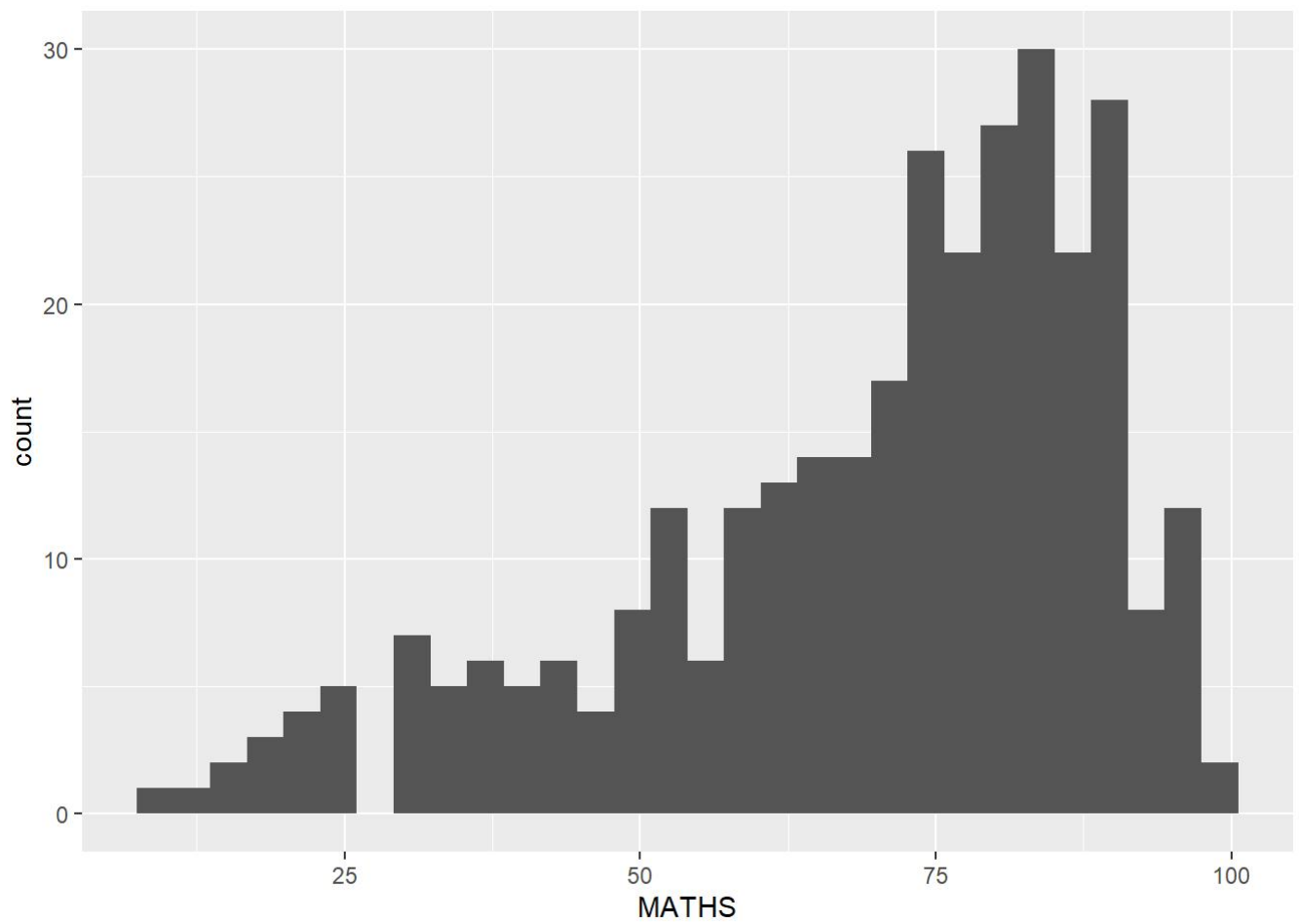
```
ggplot(data=exam_data,aes(x=MATHS))+  
  geom_dotplot(binwidth=2.5,  
               dotsize = 0.5) +  
  scale_y_continuous(NULL,  
                     breaks = NULL)
```



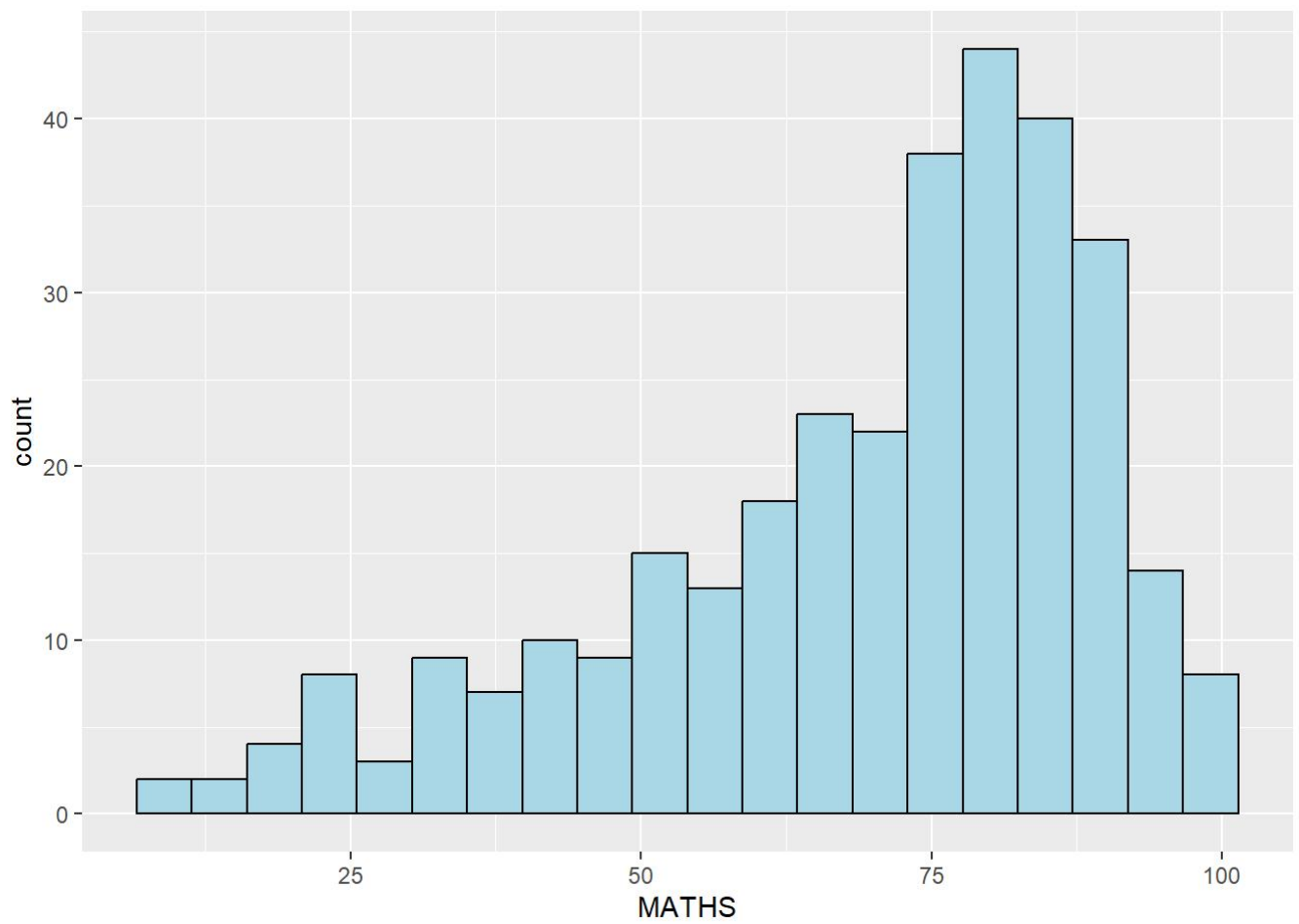
histogram

```
ggplot(data=exam_data,  
       aes(x = MATHS)) +  
  geom_histogram()
```

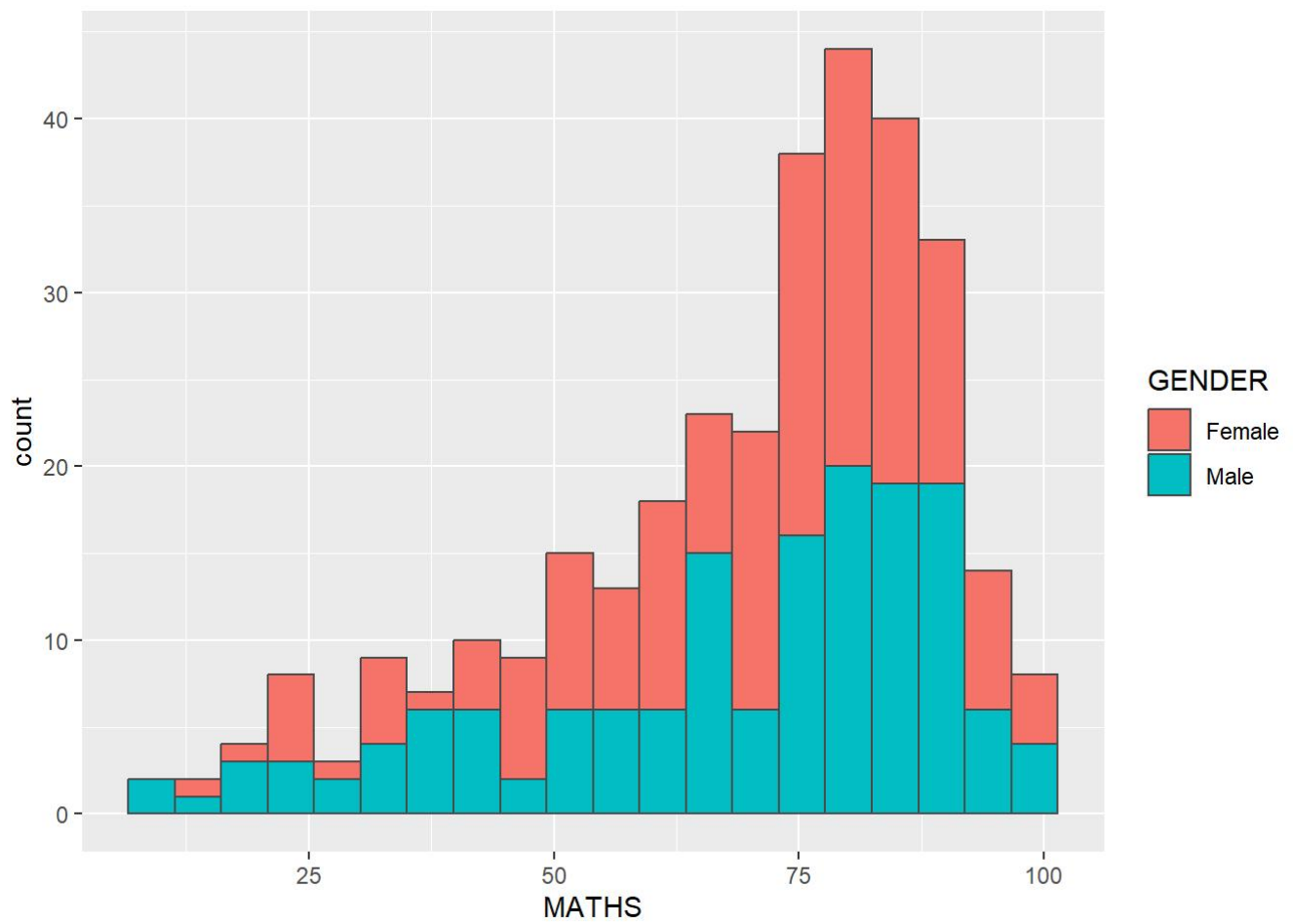
`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



```
ggplot(data=exam_data,  
       aes(x= MATHS)) +  
  geom_histogram(bins=20,  
                 color="black",  
                 fill="light blue")
```

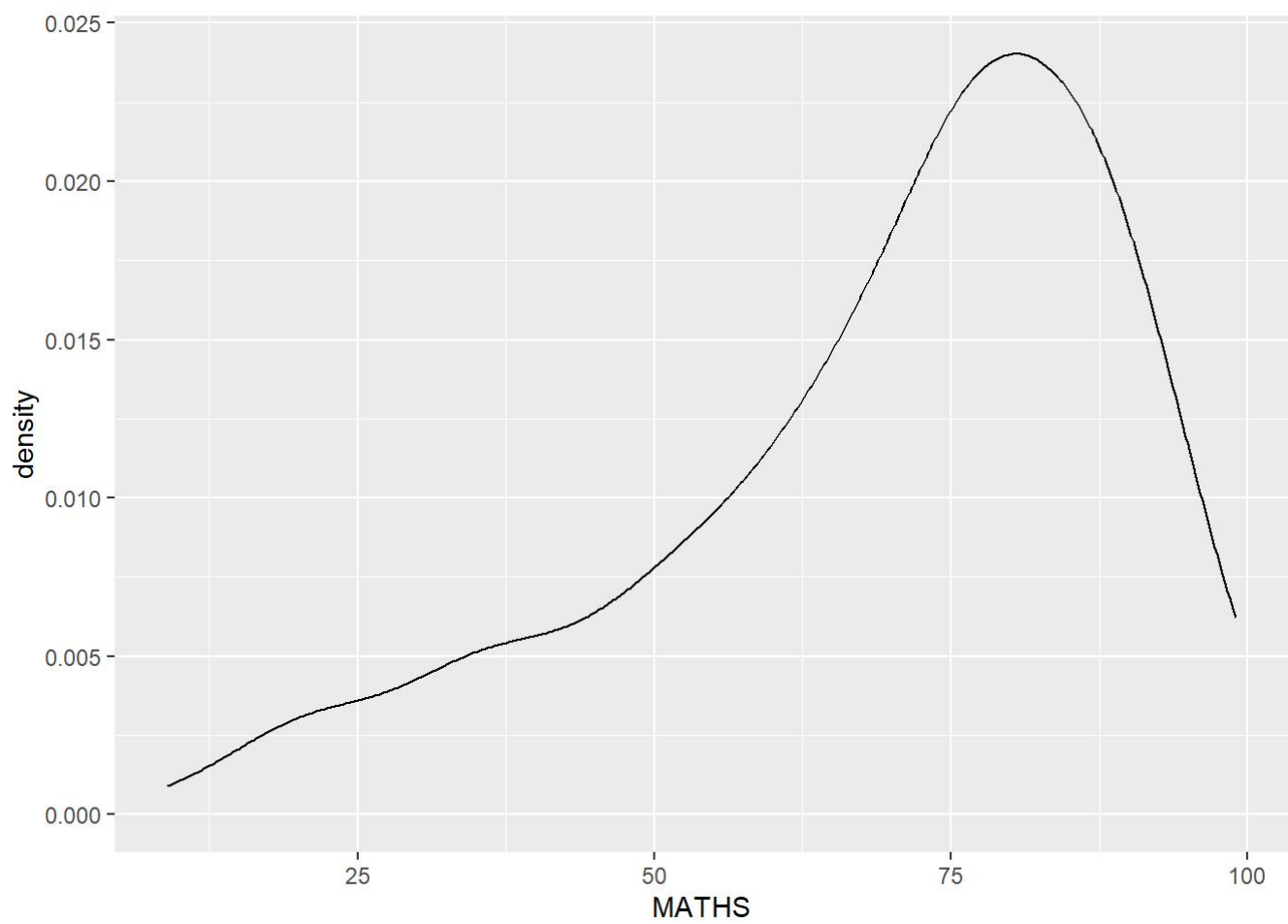


```
ggplot(data=exam_data,  
  aes(x= MATHS,  
    fill = GENDER)) +  
geom_histogram(bins=20,  
  color="grey30")
```

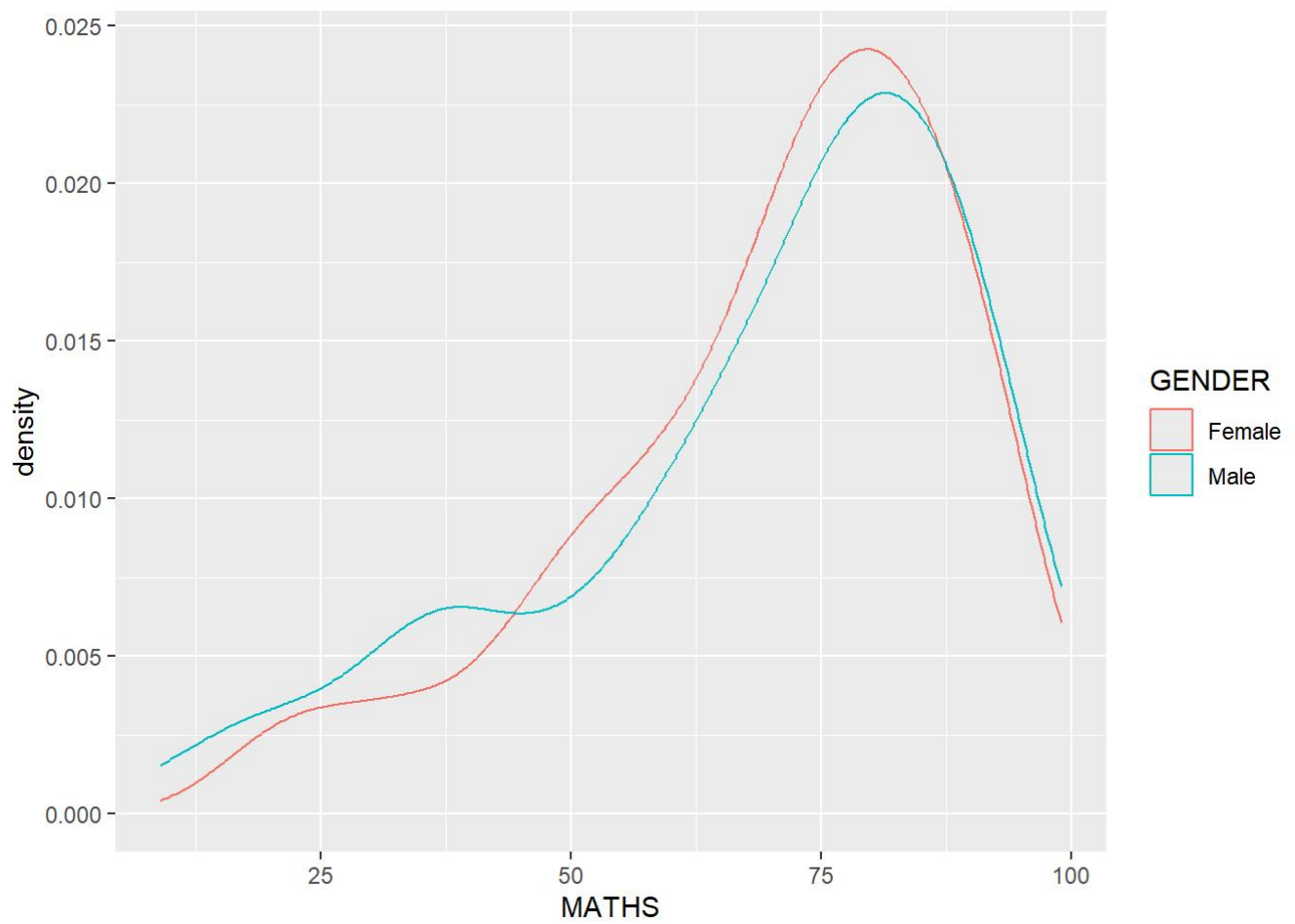


Kernel density estimate

```
ggplot(data=exam_data,  
aes(x = MATHS)) +  
geom_density()
```

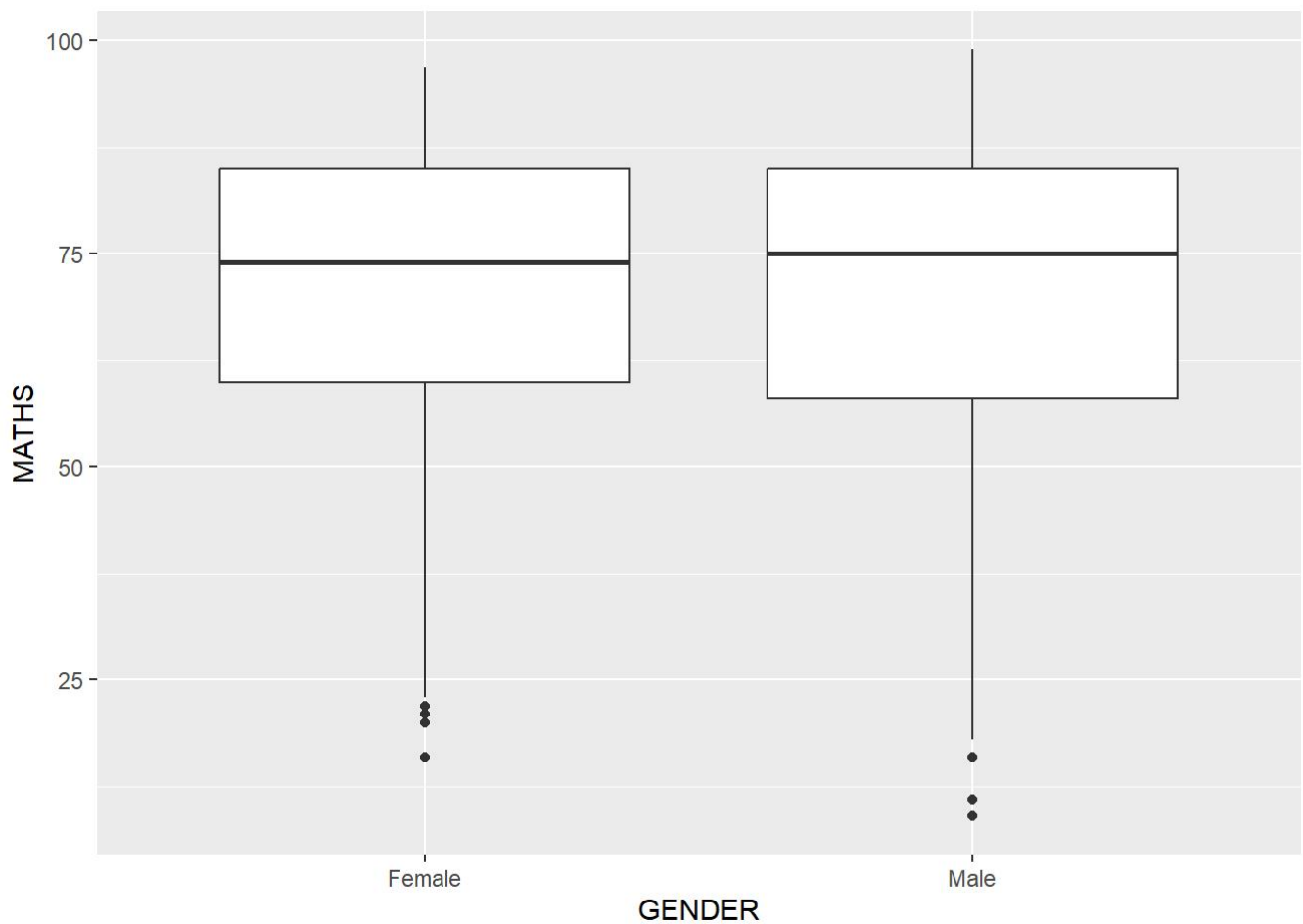


```
ggplot(data=exam_data,  
       aes(x = MATHS,  
           colour = GENDER)) +  
  geom_density()
```



Boxplot

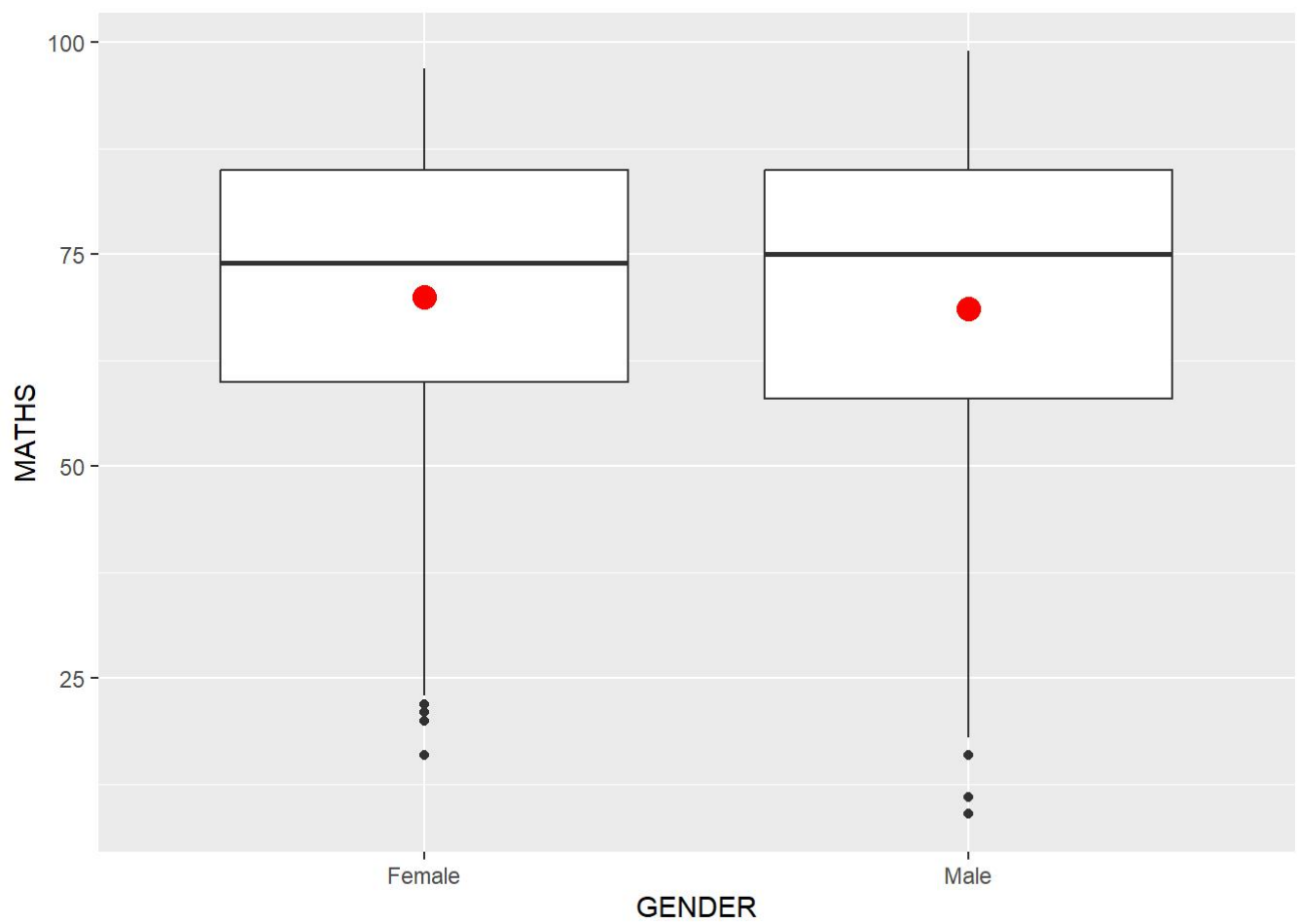
```
ggplot(data=exam_data,  
       aes(y = MATHS,  
           x= GENDER)) +  
  geom_boxplot()
```



Plotting boxplot with means

```
ggplot(data=exam_data,  
       aes(y = MATHS, x= GENDER)) +  
  geom_boxplot() +  
  stat_summary(geom = "point",  
              fun.y="mean",  
              colour = "red",  
              size=4)
```

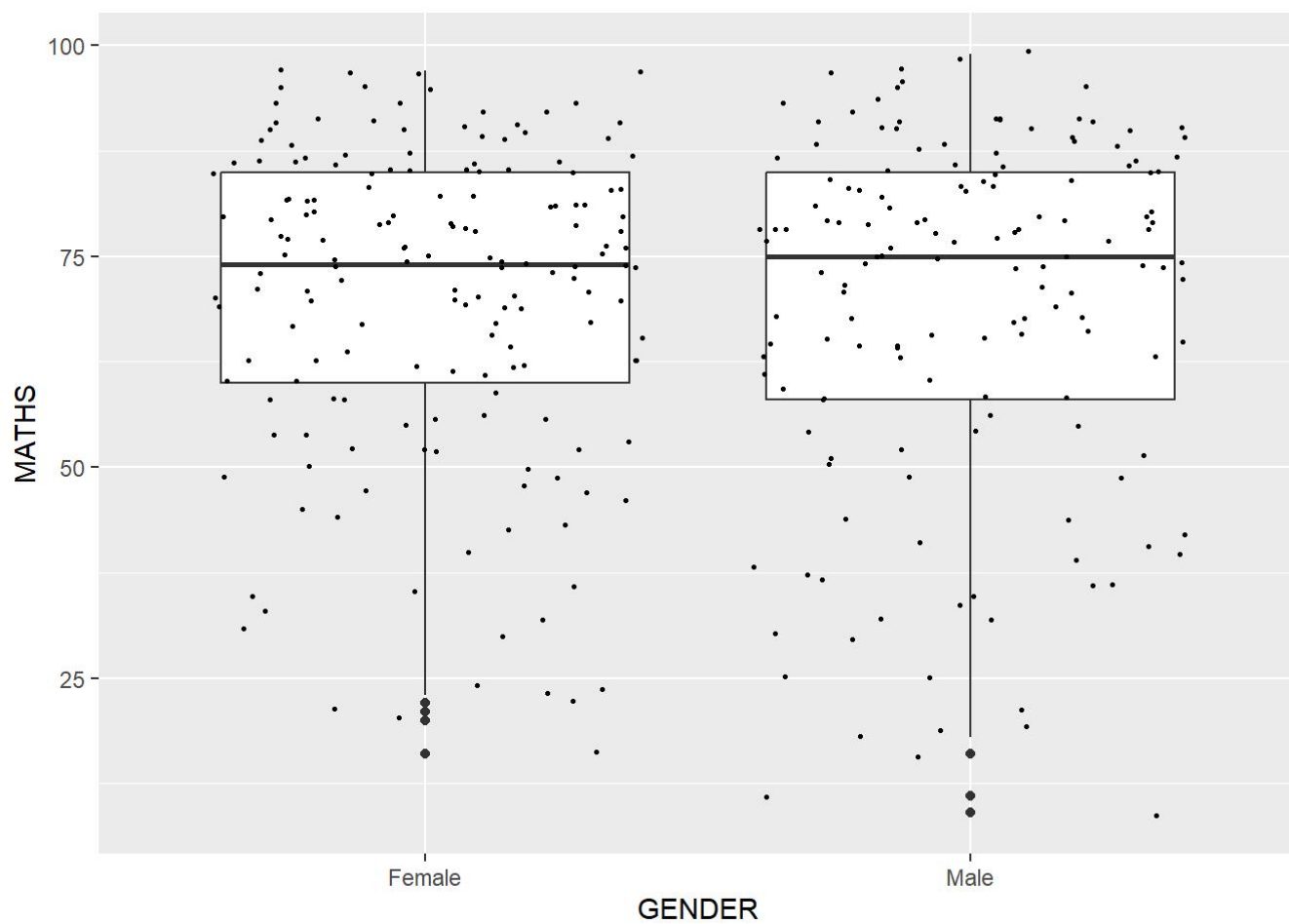
Warning: The `fun.y` argument of `stat\_summary()` is deprecated as of ggplot2 3.3.0.
i Please use the `fun` argument instead.



```
ggplot(data=exam_data,  
  aes(y = MATHS,  
      x= GENDER)) +  
  geom_boxplot(notch=TRUE)
```



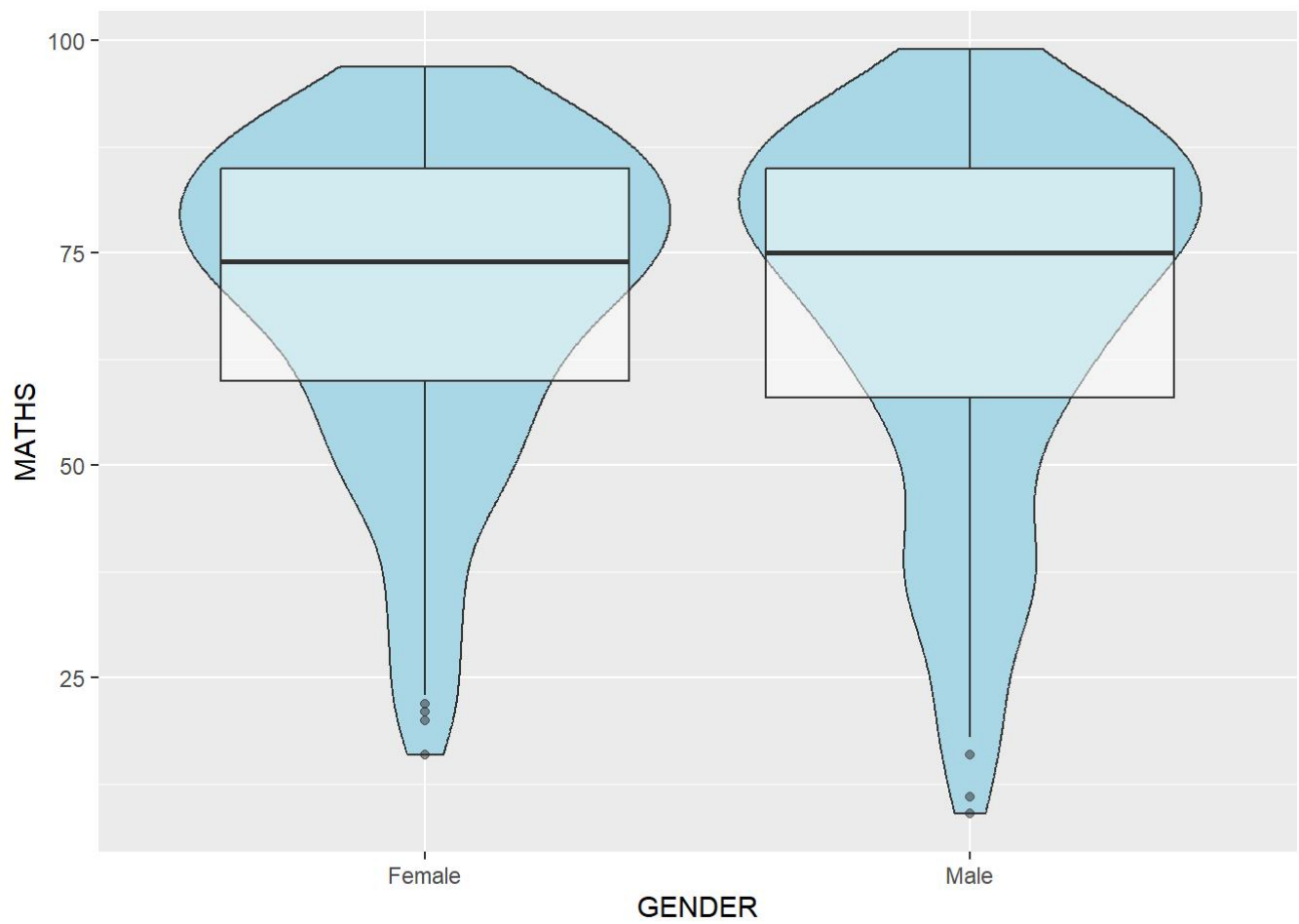
```
ggplot(data=exam_data,  
  aes(y = MATHS,  
    x= GENDER)) +  
  geom_boxplot() +  
  geom_point(position="jitter",  
    size = 0.5)
```



```
ggplot(data=exam_data,  
  aes(y = MATHS,  
    x= GENDER)) +  
  geom_violin()
```

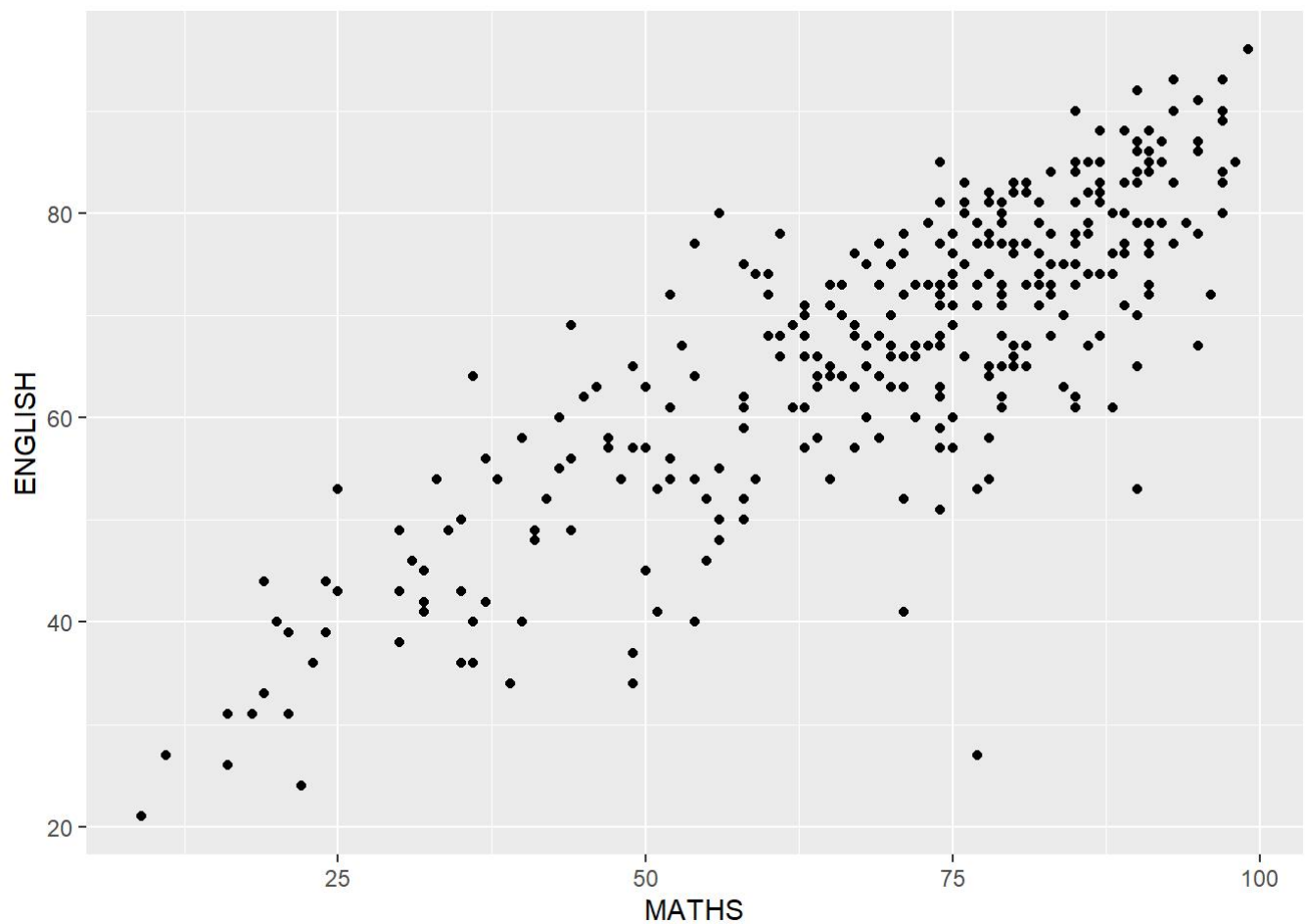


```
ggplot(data=exam_data,  
       aes(y = MATHS,  
           x= GENDER)) +  
  geom_violin(fill="light blue") +  
  geom_boxplot(alpha=0.5)
```



Scatter plot

```
ggplot(data=exam_data,  
  aes(x= MATHS,  
    y=ENGLISH)) +  
  geom_point()
```

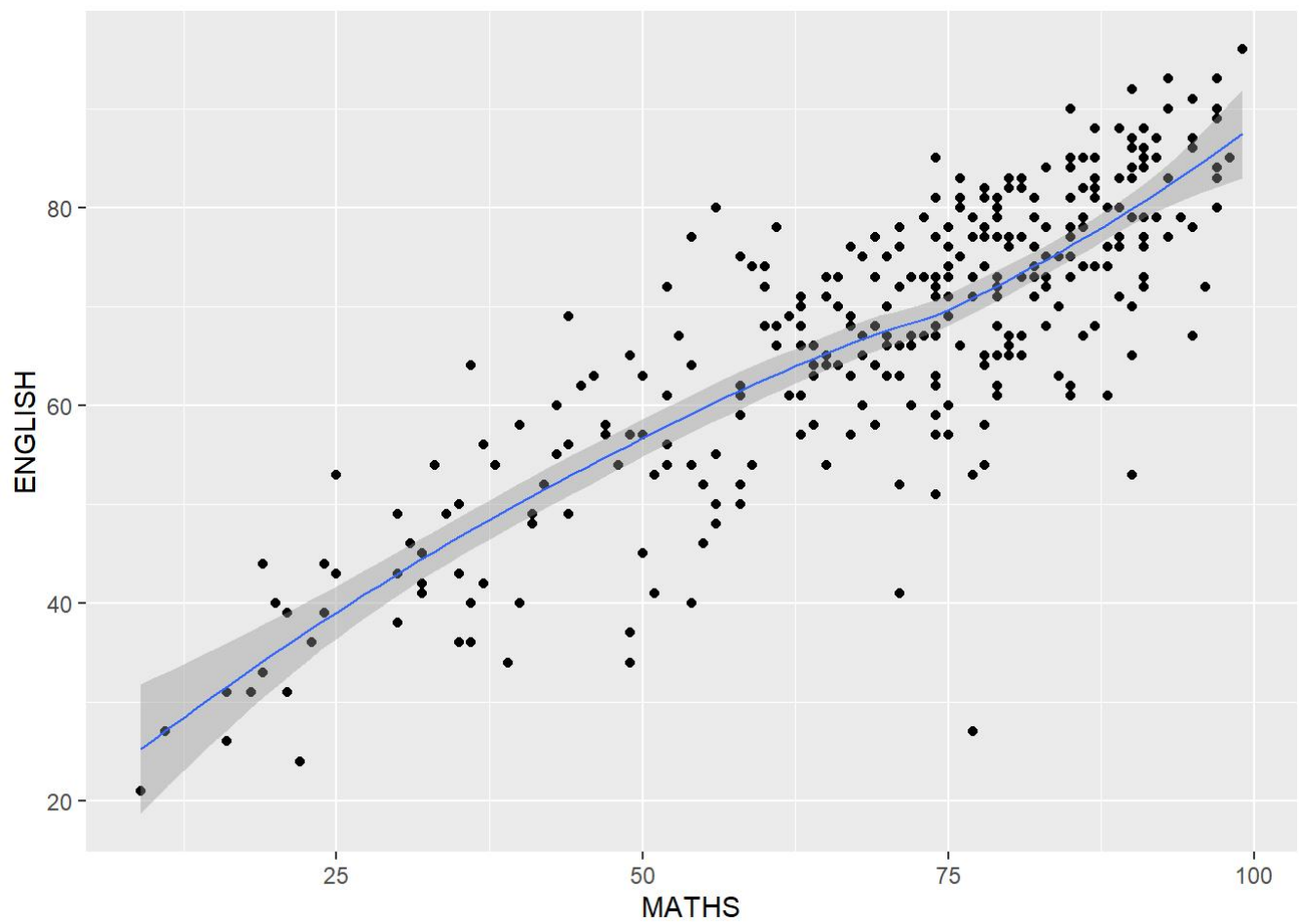


Plotting scatterplot with best fit curve

```
ggplot(data=exam_data,  
       aes(x= MATHS, y=ENGLISH)) +  
  geom_point() +  
  geom_smooth(size=0.5)
```

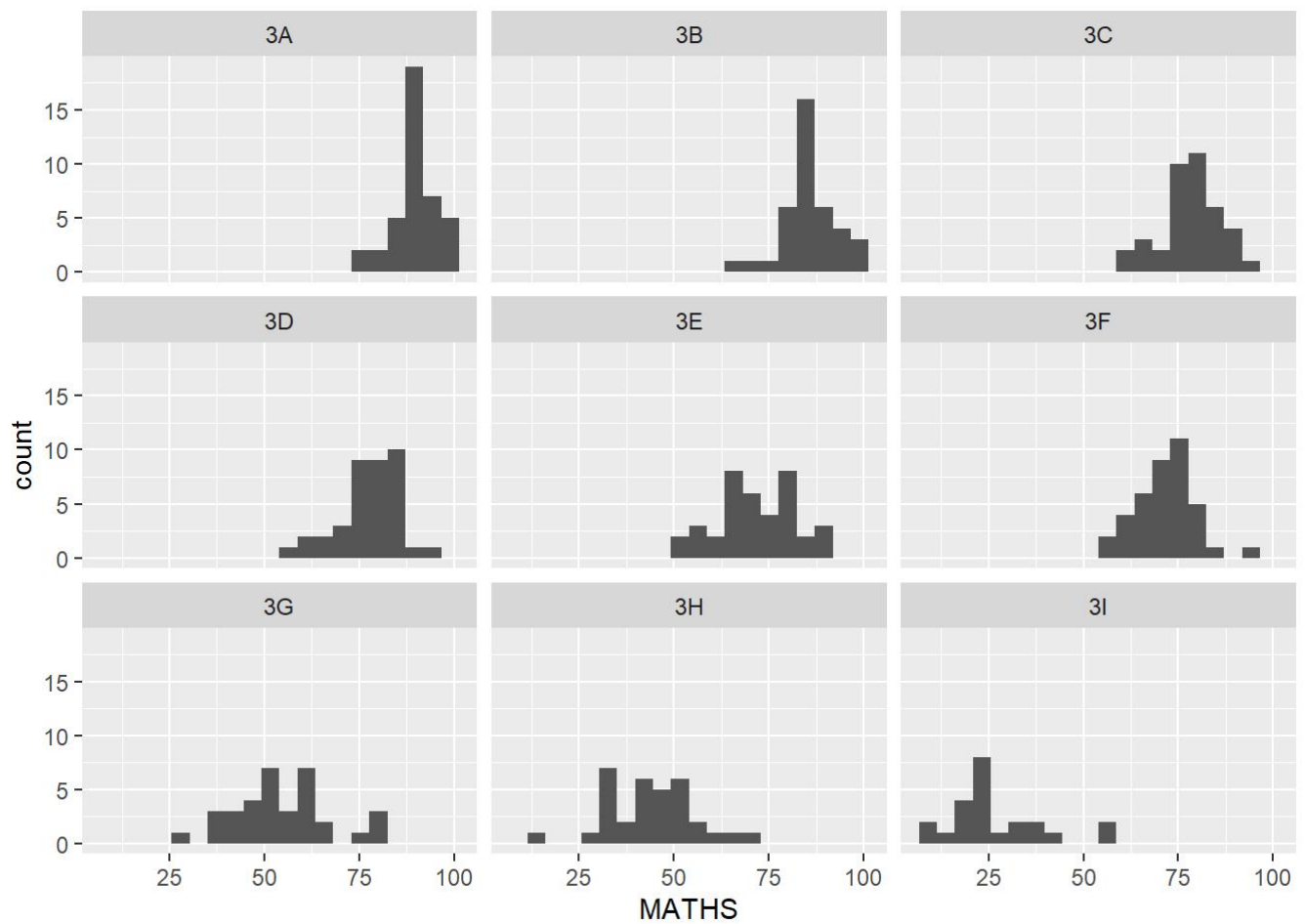
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.

`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'



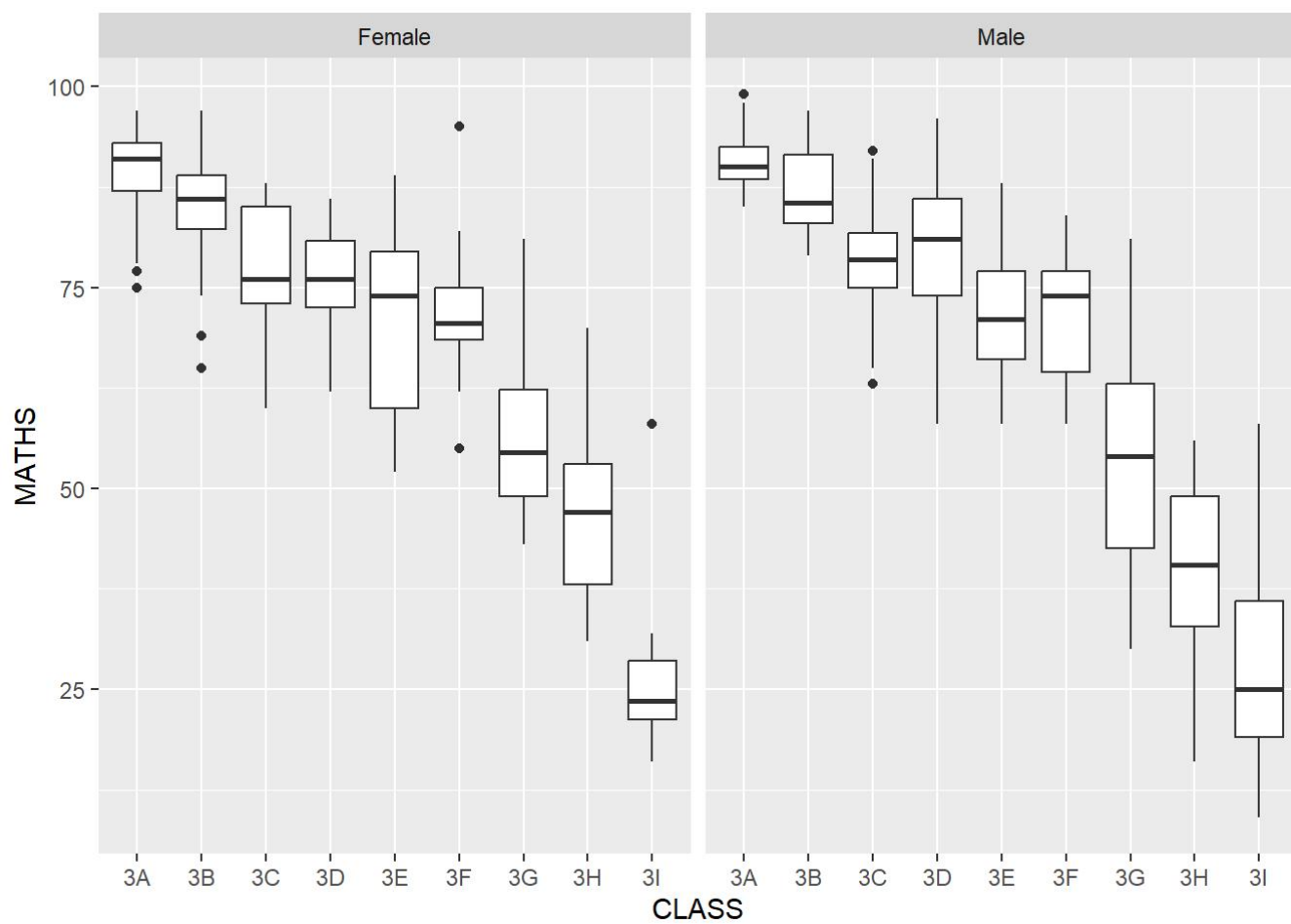
Facet

```
ggplot(data=exam_data,  
       aes(x= MATHS)) +  
  geom_histogram(bins=20) +  
  facet_wrap(~ CLASS)
```



-Box plot of math scores for classes grouped by gender

```
ggplot(data=exam_data,
       aes(y = MATHS, x= CLASS)) +
  geom_boxplot() +
  facet_grid(~ GENDER)
```



```
ggplot(data=exam_data,  
       aes(y = MATHS, x= CLASS)) +  
  geom_boxplot() +  
  facet_grid(GENDER ~.)
```

